Area in Polar Coordinates

Section 1: Sector Area Formula & Single Curves

1 Which integral formula calculates the area of a region bounded by the polar curve $r = f(\theta)$ from α to β?

a. $\frac{1}{2}\int_{\alpha}^{\beta} r^2 d\theta$

b. $\int_{\alpha}^{\beta} r^2 d\theta$

c. $\frac{1}{2}\int_{\alpha}^{\beta} r d\theta$

d. $\pi\int_{\alpha}^{\beta} r^2 d\theta$

2 Find the area enclosed by the circle $r = 4\sin\theta$ for $\theta \in [0, \pi]$.

a. 4π

b. 8π

c. 16π

d. 2π

3 What is the area enclosed by one petal of the four-petal rose $r = 2\cos(2\theta)$?

a. $\frac{\pi}{2}$

b. π

c. 2π

d. $\frac{\pi}{4}$

Section 2: Area Between Two Polar Curves

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4 Which integral represents the area inside $r = R(\theta)$ and outside $r = r(\theta)$?

a. $\frac{1}{2}\int_{\alpha}^{\beta}(R(\theta)^2 - r(\theta)^2)d\theta$

b. $\frac{1}{2}\int_{\alpha}^{\beta}(R(\theta) - r(\theta))^2 d\theta$

c. $\int_{\alpha}^{\beta}(R^2 - r^2)d\theta$

d. $\frac{1}{2}\int_{\alpha}^{\beta}R(\theta)r(\theta)d\theta$

5 Find the points of intersection of $r = 3$ and $r = 6\cos\theta$.

a. $\theta = \pm\frac{\pi}{3}$

b. $\theta = \pm\frac{\pi}{6}$

c. $\theta = \pm\frac{\pi}{4}$

d. $\theta = \frac{\pi}{2}$

6 Set up the integral for the area inside $r = 6\cos\theta$ and outside $r = 3$.

a. $\frac{1}{2}\int_{-\pi/3}^{\pi/3}(36\cos^2\theta - 9)d\theta$

b. $\frac{1}{2}\int_{-\pi/3}^{\pi/3}(6\cos\theta - 3)^2 d\theta$

c. $\int_{0}^{\pi/3}(36\cos^2\theta - 9)d\theta$

d. $\frac{1}{2}\int_{0}^{\pi/3}(6\cos\theta - 3)d\theta$

Section 3: Common Interior of Two Curves

7 How is the common interior area of two overlapping polar curves calculated when the boundary changes?

a. By finding the intersection angle and splitting the region into separate single-curve integrals

b. By subtracting the outer curve from the inner curve

c. By integrating the product of the two curves

d. By averaging the two individual areas

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8 Find the area inside both $r = 2$ and $r = 4\sin\theta$ in the first quadrant.

a. $\frac{4\pi}{3} - \sqrt{3}$

b. 2π

c. $\frac{2\pi}{3}$

d. $4\sqrt{3}$

9 Why does the pole require special care when finding intersection points of polar curves?

a. Because $r = 0$ corresponds to the same physical point (the origin) regardless of the angle θ

b. Because θ is undefined at $r = 0$

c. Because the area at the pole is infinite

d. It does not require special care

Section 4: Integration Limits & Full-Period Curves

10 What is the range of θ required to trace the limaçon $r = 2 + \cos\theta$ exactly once?

a. $[0, 2\pi]$

b. $[0, \pi]$

c. $[0, 4\pi]$

d. $[-\frac{\pi}{2}, \frac{\pi}{2}]$

11 Find the total area enclosed by the cardioid $r = 2(1 + \cos\theta)$.

a. 6π

b. 4π

c. 8π

d. 12π

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12 How many times is the circle $r = \sin(2\theta)$ traced over $[0, 2\pi]$?

a. It is a 4-petal rose traced once over $[0, 2\pi]$

b. It is a circle traced twice

c. It is traced 4 times

d. It is an ellipse

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